Report on Gaussian Elimination with Full Pivoting

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Task description

In this assignment, Gaussian elimination with **full pivoting** was implemented. The algorithm includes pivot selection from the active submatrix, row and column swaps, forward elimination, back substitution, and final reordering of the solution vector.

1. **Input matrix characteristics**

The provided 10×10 matrix was read from file and solved using full pivoting. The computed solution is very close to the vector of ones, and the residual values are very small. This shows that the implementation works correctly and stably for the assigned matrix.

2. **Outputs and brief discussion**

Why is full pivoting necessary for the prescribed real matrix?

Full pivoting is needed because it avoids using unsuitable or very small pivots during elimination. By swapping both rows and columns, the algorithm selects a better pivot from the active submatrix and improves numerical stability. This is especially important for difficult systems, where a poor pivot can strongly amplify roundoff errors.

At approximately which dimension does the Hilbert matrix become numerically problematic?

Based on figure 2, the Hilbert matrix starts to become noticeably problematic at about (n = 9).

This value (n=9) was calculated based on below condition:

```
err = maxval(abs(x - x_exact))

if (first_unstable_h == 0 .and. err > error_tol) then
  first_unstable_h = n
end if
```

Figure 1. Screenshot of a part of main.f90 script for error calculation based on threshold.

So, 9 means:
the first value of n for which $\max \ |x - x_{\text{exact}}| > 10^{-6}$.

From this point onward, the solution error grows more clearly, even though the residual remains very small. This means the system is still being satisfied numerically, but the computed solution is becoming less reliable.

Does the generalized Hilbert-type matrix behave similarly to the Hilbert matrix? Yes, the generalized Hilbert matrix shows a similar trend at first, because its error also increases as the matrix size grows. The plots (Figure 3) show that its error follows the same general pattern as the standard Hilbert matrix for smaller and medium values of (n). However, unlike the standard Hilbert case, it later reaches a numerical breakdown at (n = 47).

```
        44   183.80254803218503        5.2624571367232420E-013
        45   65.279355455052126        1.3145040611561853E-013
        46   110.04835557890667        3.7880809600210341E-013
        47                   NaN                           NaN
        48                   NaN                           NaN
        49                   NaN                           NaN
        50                   NaN                           NaN
        51                   NaN                           NaN
        52                   NaN                           NaN
        53                   NaN                           NaN
        54                   NaN                           NaN
        55                   NaN                           NaN
        56                   NaN                           NaN
        57                   NaN                           NaN
        58                   NaN                           NaN
        59                   NaN                           NaN
        60                   NaN                           NaN
        61                   NaN                           NaN
        62                   NaN                           NaN
        63                   NaN                           NaN
        64                   NaN                           NaN
        65                   NaN                           NaN
        66                   NaN                           NaN
        67                   NaN                           NaN
        68                   NaN                           NaN
        69                   NaN                           NaN
        70                   NaN                           NaN
failure summary:
Hilbert becomes unstable at n =           9
Generalized Hilbert becomes unstable at n =           9
```

Figure 2. Sample run output

Is the failure sudden, or does the error grow gradually with dimension?

For the Hilbert matrix, the deterioration is mainly gradual, since the error increases progressively with (n). For the generalized Hilbert matrix, the behavior is gradual at first, but then becomes abrupt when NaN values appear at (n = 47). Therefore, the generalized Hilbert case shows both gradual loss of accuracy and a sudden final breakdown.

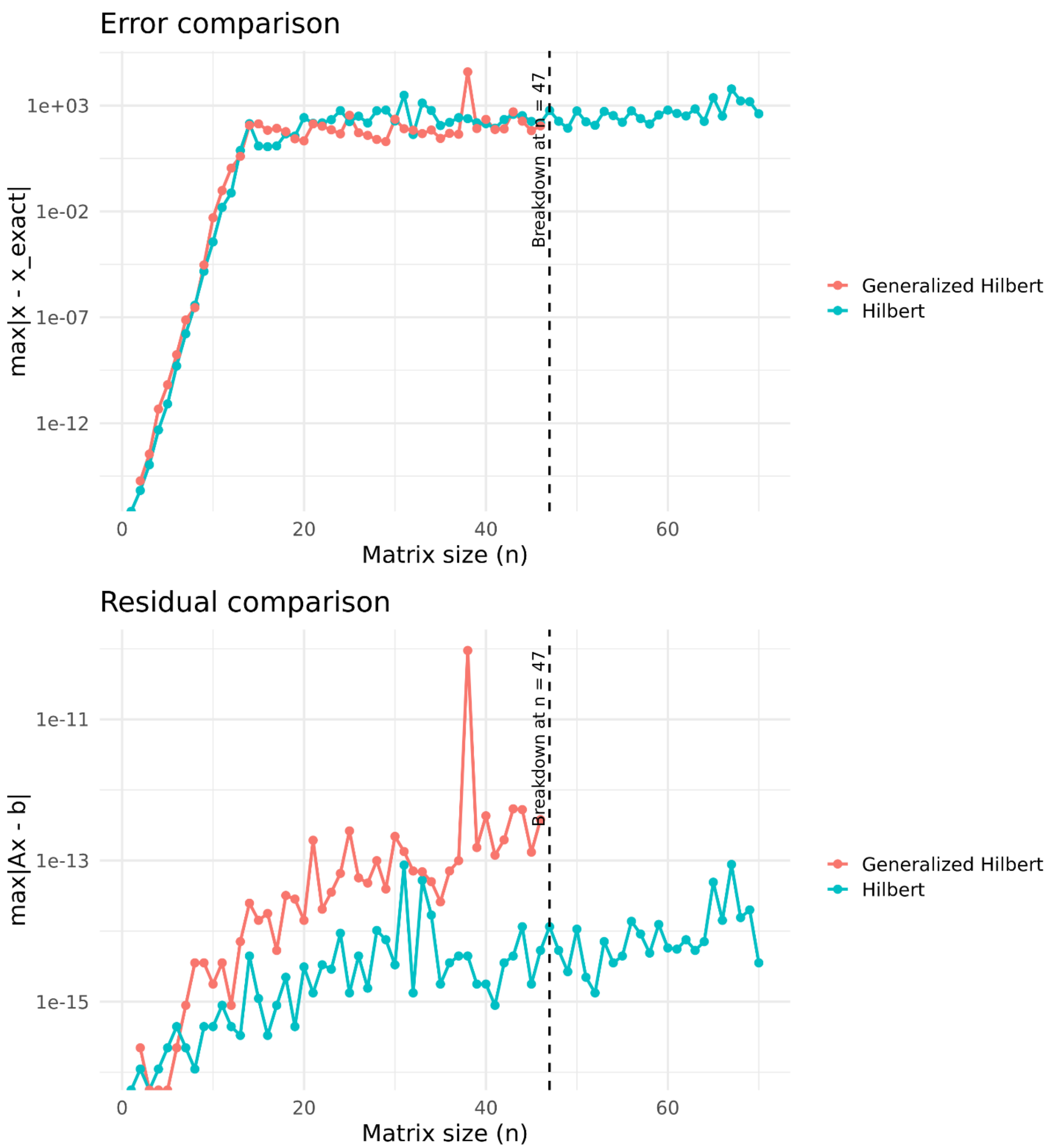

Figure 3. Output from R for error comparison and residual comparison for Hilbert and Generalized Hilbert matrices within 70 matrix size.